

receptor, and sends the results to a 24-hour medical call centre. It comes complete with a mini electrocardiogram.

"Anywhere in the world you can use it as a phone but you are also able to transfer an ECG and get a reply. If you get a normal reply, it will just be an SMS. If it's severe, you get a call 'Sir, an ambulance is on the way,'" said Dr Chow U-Jin, EPI medical chief.

Make calls without mobile towers

A team of scientists, led by Flinders University, claims to have designed a software that can run on 'off-the-shelf' mobile phones and allow them to relay calls without any cellphone tower in the area.

The technology might help during calamities by letting signals from working towers to be relayed in areas lacking signal, allowing calls in and out of the affected area. It also has the potential to dramatically improve mobile telephone coverage in many rural and remote locations where a signal is available only from limited locations in a community or with a frustratingly weak signal, say the scientists.

"Phones running our software relay calls between themselves. If even just one of those can see a cell tower, then calls can be made with any of the phones, thus sustaining communications in affected areas. A balloon is not necessary; a phone running our software at any vantage point can suffice," said Dr Paul Gardner-Stephen, who led the team.

A single remote to control all household devices

Silvan Innovation Labs has come up with a system that can connect to all the existing electric and electronic household equipment and enable users to control them by using an innovative combination of hardware and software.

There are three main components at the core of the home automation system: an infrared (IR) blaster which can be used on any remote-controlled device, a touch-screen control panel

that can switch an appliance on or off, and a software used on the panel.

The IR blaster acts as a universal remote and controls any remote-controlled equipment. Since most reputed brands have made their IR codes available openly, the software can be customised to control each device separately. The IR blaster can learn from the automation system and transmit the required IR codes directly to the equipment.

The touch-screen control panel can be mounted on a wall within the house to act as a controlling device with programmable buttons to switch on or switch off equipment at home.

The software allows notifications on a mobile phone or e-mails about instances like unusual motion detection and gas leakage.

Mobile tower in your pocket

There might have been many instances when you are on an important call and it suddenly drops due to poor network coverage, no matter how you adjust your position or fiddle with the network settings. Engineers at two UK-based firms, PicoChip and Ubiquisys, have designed gadgets that are the size of USB thumb drives and can act as a mobile network tower. The devices connect to mobile towers using the frequencies that a conventional tower uses to connect to the cellphone.

The gadgets are a shrunken form of 'femtocell'—a device that is the size of a fixed-line broadband modem which is supplied by telecom companies in areas with poor reception. PicoChip reduced the size of femtocell from its modem size to that of a USB by redesigning the main chip and reducing its power consumption to 4.5V, which is what a USB connection can supply.

"This is the world's smallest 3G base station," says Andy Gothard, PicoChip. "Once plugged into a computer, it uses a combination of listening to nearby cellular base stations and looking up the IP address of the Internet connection to work out which country it is in," adds Keith Day, vice president, Ubiquisys.

Watch 3D films on your cellphone

Watching 3D films on your cellphone will now be possible, thanks to a new technique which combines the new mobile radio standard LTE-Advanced with a video coding technique. Invented by researchers at the Fraunhofer Institute for Telecommunications, it is a special compression technique for films in especially good high-resolution HD quality that computes the films down to low data rates while maintaining quality.

What the H.246/AVC video format is to high-definition films, the Multi-



Watch 3D films on your cellphone with Fraunhofer's new technology